Journal of Clinical Epidemiology 87 (2017) 14–22

Journal of Clinical Epidemiology

GRADE guidelines 17: assessing the risk of bias associated with missing participant outcome data in a body of evidence

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Accepted 2 May 2017; Published online 18 May 2017

Abstract

Objective: To provide GRADE guidance for assessing risk of bias across an entire body of evidence consequent on missing data for systematic reviews of both binary and continuous outcomes.

Study Design and Setting: Systematic survey of published methodological research, iterative discussions, testing in systematic reviews, and feedback from the GRADE Working Group.

Results: Approaches begin with a primary meta-analysis using a complete case analysis followed by sensitivity meta-analyses imputing, in each study, data for those with missing data, and then pooling across studies. For binary outcomes, we suggest use of ''plausible worst case'' in which review authors assume that those with missing data in treatment arms have proportionally higher event rates than those followed successfully. For continuous outcomes, imputed mean values come from other studies within the systematic review and the standard deviation (SD) from the median SDs of the control arms of all studies.

Conclusions: If the results of the primary meta-analysis are robust to the most extreme assumptions viewed as plausible, one does not rate down certainty in the evidence for risk of bias due to missing participant outcome data. If the results prove not

Conflict of interest: All the authors have completed the ICMJE uniform disclosure form and declare no support from any organization for the submitted work and no financial relationships with any organizations that might have an interest in the submitted work in the previous 3 years. They declare being involved in previous publications making recommendations on the topic missing participant outcome data.

Funding: This study is part of a project on addressing missing trial participant data in systematic reviews funded by the Cochrane Collaboration. P.A.-C. was funded by a Miguel Servet research contract from the Instituto de Salud Carlos III (CP16/00137). A.G.M. was funded by a Fellowship in Guidelines Methodology by European Respiratory Society (MTF 2015-01). The funders were not involved in study design and the collection, analysis, and interpretation of data and the writing of the article and the decision to submit it for publication. The researchers are independent from funders and had full access to all the data.

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http://dx.doi.org/10.1016/j.jclinepi.2017.05.005
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Keywords: GRADE; Missing participant data; Risk of bias; Systematic reviews; Trials

1. Introduction

The extent to which risk of bias associated with missing participant outcome data (hereafter, missing data) reduce confidence in results represents a key issue for all systematic reviews [1,2]. Currently, the Cochrane Collaboration Handbook [3] focuses on determining whether individual studies are at low or high risk of bias with respect to missing data. When considering whether to rate down for risk of bias across an entire body of evidence, this approach suffers limitations. Assume, for instance, that one sets a threshold of 10% missing data for high risk of bias, and of six studies in a meta-analysis, three have no missing data and three have 12% missing data. How is one to decide whether, across the entire body of evidence, one should—or should not—rate down for risk of bias due to missing participant data?

Sensitivity meta-analyses based on different assumptions can address these issues, particularly if such analyses consider issues beyond simply the frequency of missing data, such as the event rate in the intervention and control groups, the distribution of missing data in intervention and control groups, and the reasons for missingness. The Cochrane Handbook encourages such analyses but, with respect to missing data, does not provide specific guidance regarding how to proceed.

Three prior publications have filled this gap by presenting approaches for systematic reviews of randomized trials to address missing data for binary [4] and continuous outcomes [5,6]. With some modifications, the GRADE Working Group has endorsed these approaches as GRADE guidance to assess the risk of bias associated with missing data in systematic reviews. In this article, we summarize our modified approaches, providing sufficient detail for their application, and provide several illustrative examples.

We present approaches for three situations: binary outcomes; continuous outcomes in which all studies have used the same instruments; and continuous outcomes in which studies have used different instruments to measure the same construct. In each case, the goal is to make inferences for the entire body of evidence for a particular outcome with respect to risk of bias. Within the GRADE framework, the issue is whether reviewers should rate down certainty in the evidence (quality of evidence, or confidence in evidence) for risk of bias due to missing data.

2. Development of methods

In developing our approaches, we formed a group consisting of clinical epidemiologists, methodologists, and biostatisticians, all with extensive experience in systematic reviews. We conducted a systematic survey of the literature addressing possible approaches to handling missing data when conducting a meta-analysis [7–9]. Iterative discussions among the investigators and testing our approaches in a number of systematic reviews completed the process.

The GRADE Working Group reviewed the approaches at a meeting in Vienna in October 2015, providing feedback that led to modifications from what had been previously published. The Working Group reviewed the resulting modifications, and a draft of this study, at a subsequent meeting in May 2016 and there approved the approaches as GRADE guidance.

3. Scope and definitions

This guide is for meta-analyses of trial-level data and does not address methods for meta-analyses of individual participant data that may be available to investigators. We deal only with missing data and not other elements of risk of bias in a body of evidence (e.g., allocation concealment, blinding) that systematic review authors must address.

We define participant outcome data as ‘‘missing’’ if they are unavailable to the reviewers; that is, unavailable to investigators of the primary studies, or available to the primary study investigators but not included in published reports and not provided after inquiry. A common problem when dealing with missing data is identifying whether a group of participants (e.g., those who withdrew consent or violated the protocol) have missing data or not [10–12]. Another problem is that the trial authors are sometimes not clear about how they dealt with participants missing data in their analysis (e.g., excluded them, or made assumptions) [10–13]. Before applying our approach, we recommend making all possible efforts to obtain unreported but potentially available outcome data from primary study authors, or at least understand how they dealt with missing data.

For conceptual clarity, we distinguish the issue of handling of missing participant outcome data from that of intention to treat (ITT) analysis [14]. The basic principle of ITT involves analyzing participants with available data in the arm to which they were randomized. A methodological survey found a large variation in the definition of ITT: some suggest ITT is only possible with complete follow-up; some demand imputation of missing data for an ITT analysis; and some take our position that ITT should be restricted to how one handles participants with available data, and that dealing with missing data should be treated as a separate issue [7]. Thus, what follows begins with a complete case analysis and deals with missing data as a separate issue best addressed in sensitivity analyses.

What is new?

Key findings

- When assessing risk of bias associated with participant outcome data across an entire body of evidence, we propose using a complete case analysis for the primary meta-analysis.
- When the results of the primary meta-analysis suggest a statistically significant treatment effect, conduct sensitivity meta-analyses using plausible assumptions to impute events in participants with missing outcome data in each study, and then pool across studies.

What is the implication and what should change now?

- If the results of the primary meta-analysis are robust to the most extreme plausible assumptions, one does not rate down certainty in the evidence for risk of bias due to missing participant outcome data.
- If the results are not robust to plausible assumptions, one would rate down certainty in the evidence for risk of bias.

4. Common elements of the approaches

We recommend, as do other authors who have written about the issue of missing data in the context of meta-analyses, that systematic review authors' primary analysis include only those for whom data are available (complete case analysis) [7]. An alternative is to use imputation approaches for the primary analysis, an option that is particularly attractive if investigators have strong hypotheses regarding the direction and magnitude of bias associated with missing data. Generating these alternative estimates requires considering the uncertainty associated with imputation and this consideration demands sophisticated statistical approaches. Such approaches are now available for both binary [15] and continuous variables [16].

For outcomes of putative benefit of an experimental intervention, we recommend the approaches primarily, if not exclusively, for meta-analyses in which the results suggest a statistically significant treatment effect. The purpose of the analyses is to challenge the robustness of the inference that a benefit with respect to a particular outcome does indeed exist. The approaches involve a series of progressively more stringent imputations of data in primary studies, postulating that results from participants with missing data are less favorable to the intervention than results from participants for whom the data are available. One then pools across studies to determine the impact on the point estimate and confidence interval.

For outcomes of harm (i.e., that suggest treated patients are worse of), one may challenge in a similar way the inference that apparent harm with respect to a particular outcome does indeed represent a real effect. To do so, one imputes data attributing a lower rate of adverse events in the treatment group. Alternatively, or in addition, one may attribute a higher rate of adverse events in the control group to participants with missing data than in those in whom the data are available.

In addition, one may be interested in the robustness of inferences that an intervention is not harmful. To address this issue, one would impute data suggesting a higher rate of adverse events in the treatment group. Alternatively, or in addition, one may attribute a lower rate in the control group among participants with missing data than in those in whom the data are available.

Finally, one may challenge failure to establish benefit. This would involve imputing a higher success rate in treatment group patients with missing data than in those followed and/or a lower success rate in control patients with missing data than in those followed.

5. Binary outcomes

5.1. Traditional imputations

There are many possible ways to impute missing data in individual primary studies. One might assume that all participants with missing data in either group had events, that no participants with missing data had events, or a worst-case scenario in which all participants with missing data in the intervention group suffered adverse events but none of the participants in the control group suffered such events. That worst-case scenario calculation assumes that the results of the primary analysis are suggesting the intervention reduces the incidence of the outcome of interest.

5.2. Imputations using ratios

Our suggested imputation strategy is based on making assumptions regarding the events in those with missing data as a ratio relative to those with available data in the same arm. Three such ratios have been proposed: the incidence of outcome events in participants with missing data relative to those with complete follow-up ($RI_{MPD/FU}$) [17], the informative missingness odds ratio (IMOR) [15,18,19], and the Bayesian version of the IMOR [20]. In this study, we use $RI_{MPD/FU}$ when providing illustrative examples. In positive trials, one might challenge the robustness of the results by imputing $RI_{MPD/FU} > 1$ in the intervention group and $RI_{MPD/FU} < 1$ in the control group. For instance, an event rate of 10% in participants with available data and a $RI_{MPD/FU}$ of 1.5 would result in an imputed event rate in

those with missing data of 15%. An event rate of 20% in control participants with available data with a $RI_{MPD/FU}$ of 0.5 would result in an imputed event rate in those with missing data of 10%. Similarly to the $RI_{MPD/FU}$, the IMOR describes the relationship between the unknown odds among participants with missing data and the known odds among participants with available data [15]. It differs in the use of odds instead of risks.

In trials suggesting an apparent benefit, for the sake of simplicity, we suggest a constant $RI_{MPD/FU}$ of 1.0 for control group missing participants (i.e., assume the same event rate in those with missing as those with available data). For treatment group participants with missing data, one might start with the least stringent assumptions (for instance a $RI_{MPD/FU}$ of 1.5) and repeat the meta-analysis with the associated individual primary study results. If imputed data do not materially affect the results (in particular, confidence intervals continue to exclude a null effect), one might then examine the impact of progressively more stringent but less plausible assumptions ($RI_{MPD/FU}$ of up to 3.0, or possibly 5.0).

We have used 5.0 as the most stringent but still plausible $RI_{MPD/FU}$ because we identified one study in which participants lost to follow-up were subsequently found to have had five times the rate of events than followed-up participants, but none that reported a higher ratio [21]. We refer to the meta-analysis using the plausible most stringent $RI_{MPD/FU}$ as the "plausible worst case." The reviewers should ideally select the value of the plausible most stringent $RI_{MPD/FU}$ a priori. The choice will be based on factors such as the clinical scenario (e.g., higher value of $RI_{MPD/FU}$ in a trial of cardiac transplant in which participants are more likely to have suffered a bad outcome if lost to follow-up), and the baseline prognostic profile of participants with missing participant outcome data, when reported.

To the extent that pooled estimates remain similar when making progressively more stringent assumptions (and in particular, results remain statistically significant), one would conclude that the results are robust to the missing data and, in the GRADE framework, not rate down certainty in the evidence for risk of bias. If results change materially, and particularly if statistical significance is lost, one would rate down certainty in the evidence for risk of bias due to missing data. In general, one would be more willing to rate down if significance is lost with the less stringent assumptions.

5.3. Illustrative examples

We have used this approach in several recently published Cochrane and non-Cochrane reviews [22–32]. One of these studies assessed probiotics for the prevention of *Clostridium difficile* infection (CDI) [28]. In 13 of 20 included randomized trials, data on CDI were missing for 5% to 45% of participants. We assumed that the event rate was the same among control group participants with missing data and participants that were successfully followed. For the probiotic group, we recalculated pooled treatment effects by using our assumed RI in participants with missing data compared with those who were successfully followed using the following assumptions: $RI_{MPD/FU}$ 1.5, 2.0, 3.0, and 5.0. Our results proved robust to each of the RI assumptions, and even with the 5.0 ratio, the probiotic effect remained large and the 95% CI narrow (relative risk, 0.50 [0.34–0.76]) (Appendix Fig. 1 at www.jclinepi.com).

The Appendix at www.jclinepi.com provides another example of applying the method to a benefit outcome with binary data and shows how the decision to rate down certainty in the evidence for risk of bias due to missing data can vary across outcomes within the same study (Example 2, Figures 2 and 3 in Appendix at www.jclinepi.com).

5.4. Application to harms

One could apply a similar approach to outcomes for which the results suggest harm with the experimental treatment but, in this case, impute a $RI_{MPD/FU}$ of less than 1.0 to treatment and $\geq$1.0 to control. Our suggestion, in parallel to that for benefit outcomes, is to assume $RI_{MPD/FU}$ of 1.0 for control, and a value as low as 0.20 in the intervention group. Alternatively, one could impute a $RI_{MPD/FU}$ for the intervention group and $RI_{MPD/FU}$ of $>$1.0 for the control group. Example 3 in the Appendix at www.jclinepi.com provides an illustration of use of both options.

5.5. Application to nonstatistically significant results

One could also apply the approach to determine if findings of no increase in harm are robust. This would involve the same approach as in the benefit setting: assume a $RI_{MPD/FU}$ of 1.0 in control participants with missing data and $>$1.0 in treatment group participants with missing data, possibly as high as 5.0. Again, one would examine whether results change appreciably and in particular whether previous results that were not significant become significant. Appendix Example 4 at www.jclinepi.com provides an illustration.

We have a created a freely downloadable Excel document that allows a systematic review author to determine the numerators and denominators to be used for each trial included in the meta-analysis according to the selected assumptions www.dropbox.com/s/opstwgm45qiq57k/Assumptions%20about%20MPD%20v5.xls?dl=0.

6. Binary outcomes—choosing the stringency of the imputations

Investigators using our approaches will need to decide on which extreme a $RI_{MPD/FU}$ they are willing to consider plausible. The choice will be based on factors such as the clinical scenario (e.g., higher value of $RI_{MPD/FU}$ in a trial of cardiac transplant in which participants are more likely to have suffered a bad outcome if lost to follow-up).

Another consideration will be the frequency of the event of interest. If it is infrequent (say, 5%), it may be reasonable to assume a maximum $RI_{MPD/FU}$ of 5, and thus an event rate in those with missing data of 25%. If it is frequent (say 40%), a $RI_{MPD/FU}$ of even three results in a 100% event rate in those lost. One may conclude that a rate of 100% is not plausible, in which case a maximum $RI_{MPD/FU}$ of only 2 may be appropriate.

7. Continuous outcomes—all studies using the same measure

Addressing risk of bias consequent on missing data in systematic reviews addressing continuous outcomes provides additional challenges, including the necessity of imputing both means and standard deviations (SDs). Once again, we suggest the primary meta-analysis used only participants with available outcome data (complete case). When pooled estimates are statistically significant, we suggest sensitivity meta-analyses imputing outcome data that are missing, to challenge the robustness of these pooled estimates.

To impute means, we consider five possible sources of data. In characterizing these sources, we use "best" to describe the most desirable health state (which could be a high or low score) and "worst" to describe the least desirable health state.

A. The best mean score among the intervention arms of the eligible trials.
B. The best mean score among the control arms of the eligible trials.
C. The mean score from the control arm of the trial under consideration.
D. The worst mean score among the intervention arms of the eligible trials.
E. The worst mean score among the control arms of the eligible trials.

To test the robustness of a pooled estimate showing an apparent benefit, using the five suggested sources of data mentioned previously, we recommend four imputation strategies that will almost always be progressively more stringent. Table 1 provides a matrix describing the four strategies:

- Strategy 1 uses source C for missing data in both the intervention and control arms.
- Strategy 2 uses source D for missing data in the intervention arm, and source B for missing data in the control arm.
- Strategy 3 uses source E for missing data in the intervention arm, and source B for missing data in the control arm.
- Strategy 4 uses source E for those with missing data in the intervention arm, and source A for missing data in the control arm.

We tested a number of sources of measures of variability (SDs) for the imputed data and found they yielded very similar results. We therefore suggest the simplest and most plausible source of data, the median SD in the control group of all included trials.

To generate a pooled estimate across trials using the imputed data, we suggest, for each arm in each trial, pooling the observed means and SDs of the participants with available data with the imputed means and SDs for participants with missing data using the following formulas:

$$1.\ M_{XTi} = \frac{(M_{FTi} \times n_{FTi}) + (M_{LTi} \times n_{LTi})}{n_{FTi} + n_{LTi}}$$

$$2.\ M_{XCi} = \frac{(M_{FCi} \times n_{FCi}) + (M_{LCi} \times n_{LCi})}{n_{FCi} + n_{LCi}}$$

$$3.\ SD_{XTi} = \sqrt{\frac{(n_{FTi} - 1)SD^2_{FTi} + (n_{LTi} - 1)SD^2_{LTi}}{n_{FTi} + n_{LTi} - 2}}$$

$$4.\ SD_{XCi} = \sqrt{\frac{(n_{FCi} - 1)SD^2_{FCi} + (n_{LCi} - 1)SD^2_{LCi}}{n_{FCi} + n_{LCi} - 2}}$$

$$5.\ n_{XTi} = n_{FTi} + n_{LTi}$$

$$6.\ n_{XCi} = n_{FCi} + n_{LCi}$$

where "M" represents the mean, "SD" the standard deviation, "n" the group size, "X" the combined estimates, "F" the followed-up group, "L" the lost to follow-up group, "T" the treatment group, "C" the control group, and "i" the trial.

For each study, one can then calculate the treatment effect—a mean difference—by combining means and SDs from the treatment and control arms using a fixed-effects model. One can then pool treatment effects across studies using, according to one's preference, either a standard fixed-effect or random-effects meta-analysis, to generate the mean difference across all included studies.

As was the case for the approach to binary data, if results were robust (statistical significance maintained even with the most stringent assumptions one considers plausible), one would not, within the GRADE framework, rate down certainty in the evidence for risk of bias. If statistical significance were lost for any of the more stringent plausible approach, one would rate down. Our prior papers [4–6] provide examples of use of the approach to challenging the robustness of findings of apparent benefit, as do Examples 5 and 6 in the Appendix at www.jclinepi.com.

One could apply a similar approach to harm outcomes in which the results suggest harm with the experimental treatment. In this case, the approach would involve imputing more favorable results (less harm) to those in the intervention group with missing data, and less favorable results to control group participants with missing data. The most extreme challenge would be to attribute the best mean available from either group to intervention participants with missing data, and the worst intervention group mean to control participants with missing data.

Table 1. Matrix of assumptions for participants with missing data for continuous outcomes in interventions and controls arms

Assumptions about the means of participants with missing data		Assumptions about the means of participants in intervention arm (Extreme ← Nonextreme → Extreme)				
		Source A	*Source B*	*Source C*	*Source D*	*Source E*
Assumptions about the means of participants in control arm (Extreme / Nonextreme / Extreme)	*Source A*					Strategy 4
	Source B				Strategy 2	Strategy 3
	Source C			Strategy 1		
	Source D					
	Source E					

Source A, best mean among intervention arms of included trials; Source B, best mean among the control arms of included trials; Source C, mean score from the control arm of the same trial; Source D, worst mean among the intervention arms of included trials; Source E, worst mean among control arms of included trials.

One could also apply the approach to determine if findings of no (statistically significant) increase in harm are robust. In this case, the approach would involve imputing unfavorable results (greater harm) to those in the intervention group with missing data, and favorable results to control group participants with missing data. The most extreme challenge would be to attribute the worst mean (whether it comes from intervention or control) to intervention participants with missing data, and the best mean (whether from intervention or control) to control participants with missing data. Example 7 in the Appendix at www.jclinepi.com provides an illustration.

We have a created a freely downloadable Excel document that allows a systematic review author to determine the means and SDs to be used for each trial included in the meta-analysis according to the selected assumptions per strategy: https://www.dropbox.com/s/3ie12qfwjnfwx0z/MPD%20for%20continuous%20outcomes_Template.xlsx?dl=0.

8. Continuous outcomes—studies using different measures

For certain continuous outcomes and in particular participant-important outcomes focusing on issues such as health-related quality of life (HRQL), clinical trial investigators often choose alternative measures of the same underlying construct. For example, there are at least five instruments available to measure HRQL in participants with chronic obstructive pulmonary disease (COPD) (Chronic Respiratory Questionnaire, Clinical COPD Questionnaire, Pulmonary Functional Status and Dyspnea Questionnaire, Seattle Obstructive Lung Disease Questionnaire, and the St. Georges Respiratory Questionnaire) [33]. The use of different instruments requires a modification of the methods described in the previous section.

We suggest, for this modified approach, choosing a single reference measurement instrument, converting scores from different instruments to the units of the reference instrument, and then proceeding with imputation of missing values, combining the available data with estimates from the missing data for each study, and then pooling across studies.

Alternatively, one might proceed exactly as in the example when all studies use the same instrument, but instead of natural units use the standardized mean difference (SMD). Because of limitations of the SMD both with respect to vulnerability to varying between-study heterogeneity, and its interpretability [34], we prefer to base calculations on choosing a single reference instrument as described in the following.

We suggest two key criteria when choosing the reference instrument. The first is its frequency of use, and thus its familiarity to the target audience. The second criterion is the measurement properties of the instrument. In the context of clinical trials, the key measurement properties are instrument longitudinal validity (correlations of change with other related measures), responsiveness (ability to detect important change over time, even if that change is small), and interpretability (typically, an established anchor-based minimally important difference) [35]. Details of the application of the approach follow.

Once one has chosen the reference instrument, one must convert all results into the units of that instrument. Let us say that A represents the reference instrument and B represents an alternative instrument. To convert B units to A

units, one first converts the means and SDs of the scores from instrument B to the units of instrument A [36] using the following formula:

$$M_{Ai} = (M_{Bi} - L_B) \times (R_A \div R_{Bi}) + L_A \text{ and}$$
$$SD_{Ai} = SD_{Bi} \times (R_{Ai} \div R_{Bi}),$$

where M represents the mean, L_A and L_B represent the worst possible outcome score of instrument A and B, respectively, R_A and R_B the ranges (the highest possible outcome score minus the lowest possible outcome score) for instruments A and B, respectively, and i the trial. One applies these formulas separately to the intervention group and the control group of each trial [6]. One then proceeds exactly as in the previous section using the converted score.

9. Alternative threshold for rating down for risk of bias: the context of health care guidelines

In the discussion thus far, we have suggested an approach to rating down using only one threshold: the 95% confidence interval includes a relative effect of 1.0, or an absolute difference of 0. This threshold corresponds to the *P*-value including the traditional boundary of 0.05.

This is not the only threshold one might use. Instead, one might choose the smallest effect that patients are likely to consider important and apply the approach to that threshold.

For instance, consider the outcome of prevention of a myocardial infarction. Even for an intervention associated with small burden and toxicity, patients are unlikely to choose the treatment if effects were very small (e.g., a reduction in infarction of only 1, or perhaps even 5 in 1,000). If, however, the intervention is associated with large burden and toxicity, the threshold would be much higher (10, or perhaps even 20 or more in 1,000).

Applying this logic to the latter situation and choosing a threshold of 20 in 1,000 were the boundary of the confidence interval closest to no effect to remain greater than 20 in 1,000 for even the most stringent imputation, one would not rate down certainty in the evidence for risk of bias. If, however, the confidence interval in an imputation considered plausible included the threshold of benefit of 20 in 1,000 (i.e., included reductions in infarction of less than 20 in 1,000, even if it remained above an effect of 0), one would rate down certainty in the evidence for risk of bias.

Because choosing a threshold other than no effect involves a value judgment—the choice depends on the importance placed on the target outcome (in the example myocardial infarction) and the importance placed on the burden and toxicity—this approach may be best applied in the context of a meta-analysis associated with a health care guideline. It will also be restricted to consideration of absolute rather than relative effects. We have applied this approach presented in one of our prior articles [6].

10. Dealing with limitations in reporting

Systematic review authors will find challenges when authors of primary studies fail to adequately report missing data. [10] For example, trial authors may not clearly report whether they imputed outcomes for participants with missing data. Consequently, a sensitivity analysis making imputations for participants with missing data risks double counting. Elsewhere, we have described in detail the solutions for a number of these challenges [10]. For trials in which authors do not report the frequency of missing data, we suggest using the median missing data rate from all trials included in the review. If one perceives this assumption is too stringent, alternatives include a sensitivity analysis using a missing participant data rate of zero in both arms.

For trials in which authors fail to report missing data for each study arm and report total missing data only, we suggest assuming the same rate of missing data in both intervention and control groups. For trials in which the authors report a single imputed analysis only, we suggest using the imputed results for both primary and sensitivity analyses. Reviewers should acknowledge such limitations when discussing the results of sensitivity analyses related to missing data.

11. Discussion

We have developed structured and transparent approaches to determine the extent to which missing data across an entire of evidence introduce risk of bias and thus threaten the certainty in the evidence in systematic reviews. Our approaches to binary outcomes, and to continuous data when all studies use the same outcome measure, do not require a high level of statistical sophistication, and can be carried out relatively easily in many statistical programs including RevMan which is the software developed by the Cochrane collaboration used for preparing and maintaining Cochrane Reviews including text, characteristics of studies, comparison tables, and meta-analysis. Our approach to continuous data when studies use different outcome measures begins with converting all instruments to the units of a common instrument, requires greater statistical sophistication, but is nevertheless straightforward. The approaches have received GRADE working group endorsement, and their use in any systematic review using GRADE approaches would be desirable.

The analyses that we describe are sensitivity analyses designed to facilitate inferences regarding risk of bias, rather than to generate alternative best estimates of intervention effects. Thus, the approaches do not need to deal with the uncertainty associated with the imputed values.

The approaches assume that investigators have little idea about the direction that bias as a result of missing data may take, hence the use of complete case approach in the primary meta-analysis. If investigators opt to make imputations in the primary analyses (as discussed earlier), they

should consider the uncertainty associated with imputation using the appropriate statistical approaches for both binary [15] and continuous variables [16].

Our approaches all require judgment regarding what is and is not plausible; judgments some may find arbitrary. Our approaches do, however, permit multiple progressively more stringent sensitivity analyses. This allows investigators—and users of meta-analyses—to choose the most extreme threshold that they consider plausible and then determine whether results are robust to that threshold.

We make specific suggestions for thresholds of plausibility—thresholds other than those we suggest may be more appropriate in individual meta-analyses. For instance, for continuous variables, one might not choose the most extreme results from other studies, but results adjacent to or near the extremes. Investigators concerned about confidence intervals being excessively narrow as a result of not taking into account uncertainty in imputations may choose more stringent thresholds. In general, for any particular meta-analysis, those who consider extremes more plausible will be more likely to rate down the certainty of the evidence for risk of bias due to missing data than those who do not.

Deciding on specific imputation strategies also entailed some degree of arbitrariness: we opted, where possible, for simplicity. For example, to address apparently, beneficial treatment effects in binary outcomes we suggest, for the control group, assuming that event rates in missing participants do not differ from those in participants with complete data. Thus, the only variation is the increase in event rates imputed to missing data, relative to those with complete data, in the intervention group. It is possible, of course, for investigators to reasonably deviate from our guidance and to also vary control group event rates imputed to control groups. Systematic review authors might consider similar reasonable alternatives regarding our suggestions for how to deal with harm outcomes, and with continuous variables.

In our presentation, we have focused on rating down certainty in the evidence for risk of bias only when meta-analyses that include plausible imputations for missing data result in loss of statistical significance. We have also pointed out, however, that one could be even more stringent: one could rate down if the boundary of the confidence interval closest to no effect includes a threshold of patient importance.

In summary, this GRADE guidance includes structured, transparent, and relatively easily implementable strategies to determine whether the extent of missing data warrants rating down the certainty in a body of evidence for a particular outcome for risk of bias. Ongoing work involves examining the impact of the approaches on a large sample of meta-analyses and may inform future updates of this guidance [37].

Acknowledgments

Authors' contributions: G.H.G., S.E., P.A.-C., B.C.J., R.A.M., S.D.W., and E.A.A. contributed to the conception and design of the study. G.H.G., S.E., P.A.-C., B.C.J., A.G.M., M.B., R.A.M., X.S., S.D.W., D.H.-A., I.N., L.A.K., A.I., J.M., H.J.S., and E.A.A. contributed to the analysis and interpretation of the data. S.D.W. and D.H.-A. contributed to the statistical expertise. G.H.G. contributed to drafting of the article. G.H.G., S.E., P.A.-C., B.C.J., A.G.M., M.B., R.A.M., X.S., S.D.W., D.H.-A., I.N., L.A.K., A.I., J.M., H.J.S., and E.A.A. contributed to the critical revision of the article for important intellectual content. G.H.G., S.E., P.A.-C., B.C.J., A.G.M., M.B., R.A.M., X.S., S.D.W., D.H.-A., I.N., L.A.K., A.I., J.M., H.J.S., and E.A.A. contributed to the final approval of the article.

Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jclinepi.2017.05.005.

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